estimated to have an upper bound of $375,000.[3030] Incremental audit fees are estimated to have an upper bound of $23,000 for all years.[3031]

We emphasize that there could be a considerable range in actual compliance costs given that not all costs listed above will apply to all registrants or during all measurement periods. Depending on the registrant, annual compliance costs (averaged over the first ten years of compliance) could range from less than $197,000 to over $739,000.[3032] A registrant's compliance costs may be at the lower end of the cost range if, for example, it does not conduct scenario analysis, does not have material Scope 1 and 2 emissions, has no climate-related target or goal, and has no applicable expenditures or financial statement impacts that require disclosure, thereby avoiding the corresponding costs of the aforementioned disclosure items. However, this registrant may have exposures to material climate risks that necessitate governance disclosure; disclosure regarding climate-related risks that have material impacts on strategy, business model, and outlook; and risk management disclosure. In this case, the cost of these required disclosures—estimated to be $327,000 in the first year of compliance and $183,000 annually in subsequent years[3033]—would comprise the full compliance cost of the final rules. This corresponds with an average annual compliance cost of $197,000 (rounded to the nearest $1,000) over the first ten years of compliance.[3034] Incremental compliance costs would be even lower for

---

3030 *See* section IV.C.3.b.v.

3031 *Id*.

3032 Registrants will incur compliance costs for different disclosure items at different times due to applicable phase in periods. For ease of comprehension and comparability, these estimates are presented as the average annual compliance cost over the first ten years of compliance. *See infra* notes 3034 and 3036 for additional details.

3033 *See* section IV.C.3.b.i.

3034 ($327,000 + $183,000*(9 years))/10 = $197,400.

registrants that already provide these disclosures (either voluntarily or as required by other laws or jurisdictions).[3035]

At the upper end of the cost range, for example, there may be other registrants for which all estimated compliance costs apply. In this example, these registrants could incur an estimated $872,000[3036] in the first year of compliance and lower annual costs in subsequent years. After the respective phase in periods, these registrants would incur additional costs for GHG emissions disclosure, limited assurance, and subsequently reasonable assurance (assuming the registrant is an LAF). This registrant would incur an average annual compliance cost of $739,000 (rounded to the nearest $1,000) over the first ten years of compliance.[3037] These examples highlight the potential range in compliance costs depending on a given registrant's circumstances, including (but not limited to) industry, size, existing climate-related disclosure practices, and whether the registrant's climate-risk exposure exceeds applicable materiality thresholds for disclosure.

Regarding assessing materiality to determine whether disclosure is required under the final rules, we acknowledge that some registrants may need to expend resources to first determine whether particular disclosure items are material, even in cases where registrants ultimately determine they do not need to make disclosure. While commenters provided estimates of the overall costs of measuring and assessing GHG emissions and making disclosure

---

3035 *See supra* section IV.A for a discussion on existing laws (domestic and foreign) that elicit similar disclosures and current market practices with respect to climate-related disclosures. *See also infra* section IV.C.3.c.

3036 $327,000 (governance disclosure; disclosure regarding climate-related risks that have material impacts on strategy, business model, and outlook; and risk management disclosure) + $12,000 (reporting cost of scenario analysis) + $10,000 (reporting cost of target or goal) + $500,000 (disclosures related to amendments to Regulation S-X, upper bound) + $23,000 (audit fees, upper bound) = $872,000.

3037 Total compliance costs are calculated each year for the first ten years of compliance, taking into account the various disclosure items and their respective phase in periods. The average of these annual costs is $738,700.

under TCFD disclosure frameworks, they did not provide a level of detail that would enable us to reliably disaggregate the materiality determination from the costs of disclosure more broadly. We also note that the cost of such a determination could vary depending on the registrant's facts and circumstances and may in some cases be de minimis. While we have not provided a standalone cost estimate of making such materiality determinations, our estimates of the costs of governance disclosure, disclosure regarding the impacts of climate-related risks on strategy, business model, and outlook, and risk management disclosure begin with TCFD disclosure as a starting point.[3038] Thus, to the extent that a materiality or similar assessment is included in TCFD disclosure, this cost is reflected in the Commission's compliance cost estimates with respect to the above disclosure items.

Moreover, the above estimates are conditional upon several factors. First, they depend on the sample of sources and commenters that voluntarily provided relevant cost information.[3039] To the extent that this sample is not representative of the broad set of affected registrants, the resulting estimates may similarly be less representative. In addition to company size and industry, another relevant factor may be the decision to engage third-party advisory services. Some registrants may determine that engaging such advisory services will better position them to comply with the final rules, while others may decide to use in-house resources.[3040] The above estimates incorporate information on both internal costs (*e.g.*, employee hours) and external costs

---

[3038] See section IV.C.3.b.i.

[3039] *See supra* section IV.C.3.a.

[3040] For example, registrants that are required to disclose emissions may be more likely to rely on external services. Registrants facing climate-related risks that are complex or a myriad may also be more likely to engage third party services. We emphasize that the final rules impose no requirement with respect to the use of third-party services and that registrants are free to decide how best to meet compliance based on their specific circumstances.

(*e.g.*, hiring third parties or consultants), as provided by comment letters and other sources.[3041] Second, several analytical assumptions were incorporated in the estimation process. While we endeavored to apply them consistently and in a conservative manner throughout the analysis, actual compliance costs may differ to the extent that these assumptions do not reflect a given registrant's specific circumstances.

The above compliance cost estimates exhibit certain features that may make them conservative. First, the cost estimates from comment letters and other sources, which serve as inputs in our cost estimation process, are almost all from large-cap companies. To the extent that compliance costs increase with company size, smaller registrants can expect lower costs.[3042] Furthermore, there are numerous instances in which analytical assumptions were required due to insufficient information from the source material. Wherever possible, assumptions that tend to overstate actual costs were chosen over those that would tend to understate them. Certain registrants may nonetheless incur costs that exceed our estimates. However, we believe that due to the nature of our cost estimation process, the majority of registrants will incur costs that do not exceed our estimates. Furthermore, our estimates assume registrants have no pre-existing climate-related disclosure practices. As a result, those that already provide disclosures that meet some of the final rules' requirements will face lower incremental costs.[3043]

[3041] Some commenters provided TCFD disclosure costs and separate costs for sustainability consultants. *See, e.g.,* letters from Soc. Corp Gov (June 11, 2021, and June 17, 2022). However, the latter were often not explicitly tied to TCFD, but rather associated with sustainability reports or other disclosures and activities not necessarily required by the final rules. In these cases, we only used the TCFD disclosure costs due to their direct relevance while omitting the cost of sustainability consultants as we could not reliably determine what portion were directly attributable to the TCFD and the provisions of interest. For GHG emissions, some companies' estimates included both internal and external costs, some mentioned the use of external costs but did not provide dollar estimates, while others did not engage external services at all. We have incorporated all available information to the extent possible in our estimation process.

[3042] Nevertheless, we recognize that in some cases, certain components of compliance costs may not vary with size and may be higher in proportional terms for smaller registrants.

[3043] *See* section IV.C.3.c.

We recognize that some comment letters in response to the proposed rules contained compliance cost estimates that significantly exceed the Commission's estimates of the final rules.[3044] We reiterate that this discrepancy is likely attributable to a number of changes from the proposed rules that reduce compliance costs. For example, the final rules do not require Scope 3 emissions reporting and have less burdensome requirements with respect to the amendments to Regulation S-X, thereby resulting in reduced compliance costs.

Our compliance cost estimation process consists of five elements. First, we estimate the aggregate costs of complying with three specific provisions that have similar counterparts within the TCFD framework: governance disclosure; disclosure regarding climate-related risks that have material impacts on strategy, business model, and outlook; and risk management disclosure. Second, we estimate the cost of assessing and disclosing Scope 1 and 2 emissions. Third, we estimate the cost of obtaining third-party assurance for GHG emissions disclosures. Fourth, we estimate the reporting costs of scenario analysis and targets and goals. Fifth, we estimate the costs associated with complying with the amendments to Regulation S-X and incremental audit costs. We proceed with a review of each element that describes how we arrived at the above compliance cost estimates.

i. Cost estimates of Governance Disclosure; Disclosure Regarding Impacts of Climate-Related Risks on Strategy, Business Model, and Outlook; and Risk Management Disclosure

We begin by reviewing estimates from commenters and other sources with respect to the costs of TCFD disclosure with the objective of informing our assessment on the costs of similar provisions of the final rules. Specifically, these provisions of interest include governance

---

[3044] *See* section IV.C.3.c.

disclosure; disclosure regarding climate-related risks that have material impacts on strategy, business model, and outlook; and risk management disclosure. We begin by focusing on these specific provisions separate from other components (*e.g.,* GHG emissions measurement or targets and goals) because these other components are not necessarily required in all circumstances or by all registrants.

In many cases, however, commenters provided one aggregate cost for their TCFD disclosure that also included the costs of GHG emissions measurement or target and goals-related activities. Without a breakdown of component costs, we face challenges in isolating the costs of the relevant provisions. Moreover, some commenters provided only a single aggregate cost that, in addition to their TCFD disclosure, includes several other components not required by the final rules,[3045] which poses similar challenges in separately estimating the component costs.

To account for these challenges, we used an approach that takes these aggregate cost estimates and applies adjustments derived from specific estimates from other sources, allowing us to obtain a more targeted "adjusted cost." For example, some commenters provided their cost estimates specifically for measuring emissions, from which we can determine the median reported emissions-measurement cost. Thus, if a given commenter provided an aggregate cost of TCFD disclosure that includes the measuring of emissions, we applied an adjustment (i.e., subtracted the median reported emissions-measurement cost), which results in an adjusted cost estimate for the remaining portion of TCFD disclosures (i.e., the provisions of interest). We applied similar adjustments throughout the analysis, as described in detail below.

[3045] For example, an anonymous large-cap company "noted that combined costs for producing its first TCFD, SASB, and GRI disclosures were between $200,00 and $350,000." *See supra* note 2987.

While this approach can help us arrive at more granular cost estimates, we also recognize its limitations. Primarily, the median reported cost of a given component may be different from the actual cost incurred by a specific registrant (due to differences in company size, industry, climate reporting practices, or other factors) such that applying the adjustment may not yield a true representation of that registrant's cost breakdown. However, we believe this issue is mitigated to some extent because almost all estimates used in this analysis are from large cap companies and thus of relatively comparable size. Furthermore, while a given cost adjustment may be overstated for some registrants and understated for others, these discrepancies should partially offset each other when we subsequently take the median[3046] of the resulting adjusted costs.

Table 8 presents an overview of the cost estimation methodology with respect to the provisions of interest. Column (1) specifies the commenter or other public source that contains cost estimates specific to TCFD disclosures. Some sources contained costs for multiple, anonymous companies. Where applicable, these company descriptions are provided in Column (2). Column (3) shows the ongoing costs of TCFD disclosures before cost adjustments are applied. Some costs are taken directly from the source, whereas in other cases, specific assumptions and calculations are applied to obtain an estimate (see table footnotes for details). For example, if a source provided estimates in the form of FTEs or burden hours, we converted them to dollars according to hourly cost estimates consistent with the PRA.[3047] Some sources only provided an initial cost (i.e., first-year startup cost) without providing ongoing, annual costs.

[3046] Throughout the cost estimation process, we use medians instead of means since the former is less sensitive to outliers.

[3047] The PRA assumes that internal burden hours cost $444/hour, while external burden hours cost $600/hour. *See* section V.

In these cases, we estimate the ongoing cost by applying a percentage reduction derived from other sources. Furthermore, because the CDP questionnaire exhibits full alignment with the TCFD recommendations,[3048] we also included estimates for responding to the CDP questionnaire, from which we estimated the equivalent cost for TCFD disclosures by applying a conversion factor.[3049]

We determined that some of the costs in Column (3) include the costs of setting targets and goals or measuring GHG emissions, as indicated in Columns (4) and (5), respectively. Where applicable, these costs are subtracted from Column (3) to obtain the adjusted cost in Column (6), which represents the aggregate, annual ongoing cost estimate for provisions of interest: governance disclosure; disclosure regarding climate-related risks that have material impacts on strategy, business model, and outlook; and risk management disclosure.

3048 For information on how the CDP questionnaire is fully aligned with the TCFD, *see* CDP, How CDP is aligned to the TCFD, *supra* note 52.

3049 Two companies referenced in comment letters noted that it takes a designated number of staff four months to complete the CDP questionnaire and nine months to complete TCFD disclosures. Based on these estimates, we incorporate the assumption that the CDP-to-TCFD cost ratio is 4 to 9 ("4-to-9 ratio"). *See* letters from Soc. Corp. Gov. (June 11, 2021 and June 17, 2022).

Table 8. Cost Estimates of Governance Disclosure; Disclosure Regarding Climate-Related Risks that have Material Impacts on Strategy, Business Model, and Outlook; and Risk Management Disclosure

Commenter or Source	**Type of Company** (if specified)	**Ongoing Cost of TCFD disclosures** (pre-adjustment)	**Adjustment: setting targets and goals** ($54,015) [1]	**Adjustment: Scope 1, Scope 2, and some Scope 3 emissions** ($79,236) [2]	**Adjusted cost** [3]
(1)	(2)	(3)	(4)	(5)	(6)
Society for Corporate Governance (June 11, 2021)	Financial Services company	$1,918,080 [4]	Included	Included	$1,784,829
	Energy company	$8,524,800 [5]		Included	$8,445,564
	Comm. Services company	$865,385 [6]	Included	Included	$506,862
Society for Corporate Governance (June 17, 2022)	Company 5	$360,000 [7]		Included	$280,764
	Company 6	$2,237,760 [8]	Included	Included	$2,104,509
FCA rule		$86,270 [9]			$86,270
BEIS rule		$40,694 [10]			$40,694
The Climate Risk Disclosure Lab	Large-cap financial institution	$56,000 [11]			$56,000
	Mid-cap company	$5,600 [12]			$5,600
	Large-cap company	$63,000 [13]			$63,000
American Exploration and Production Council		$280,000 [14]			$280,000

S&P Global	$56,000 [15]	$56,000
Median		**$183,135** [16]

1. The adjustment factor for setting targets and goals is $54,015, which is determined by relevant cost estimates presented in Table 9.
2. The adjustment factor for assessing Scope 1, Scope 2, and some Scope 3 emissions is $79,236, which is determined by relevant cost estimates presented in Table 10.
3. The adjusted cost is calculated as Column (3) minus adjustment factors where applicable, as indicated by Columns (4) and (5). If Column (4) indicates "Included," then $54,015 is subtracted from Column (3). Similarly, if Column (5) indicates "Included," then $79,236 is subtracted from Column (3). The net result is the "adjusted cost," presented in Column (6).
4. *See* letter from Soc. Corp. Gov (June 11, 2021). This company reported that three FTEs "plus others" spend nine months for TCFD reporting. (3 FTEs)*(40 hrs/wk)*(36 wks)*($444/hr) = $1,918,080. The source does not specify how many hours are contributed by the "others," thus the estimated cost may be understated.
5. *See id.* This company reported TCFD-aligned reporting process involved 40 people from the company and took six months of nearly full-time participation by 20 core team members. (20 FTEs)*(40 hrs/wk)*(24 wks)*($444/hr) = $8,524,800. The source does not specify how many hours are contributed by those outside of the 20 core members, thus the estimated cost may be understated.
6. *See id.* This company reported spending $1.25 million on both CDP and TCFD disclosures, in addition to several other components. We first estimate the TCFD component by applying the 4-to-9 ratio. ($1.25 million)*(9/13) = $865,385.
7. *See* letter from Soc. Corp. Gov (June 17, 2022). This company reported spending "$160,000 for CDP and other climate-related surveys, including supply chain surveys." To be conservative, we assume that the $160,000 is the cost for CDP only, then apply the 4-to-9 ratio. $160,000*(9/4) = $360,000.
8. *See id.* This company reported that "two employees focus on climate change, including disclosure, and 1.5 employees focus on sustainability reporting overall," spending nine months on its TCFD report. (3.5 FTEs)*(40 hrs/wk)*(36 wks)*($444/hr) = $2,237,760.
9. *See supra* note2995. This is the ongoing cost of "coordination of disclosure inputs across functions" (£69,747 for larger issuers), which is in line with the TCFD disclosure categories of Governance, Strategy, and Risk Management. This cost is converted to USD based on the 2022 average exchange rate. (£69,747)*(1.2369 USD/GBP) = $86,270. This reflects a 56% reduction from initial to ongoing costs, which we consider in determining the appropriate percentage reduction in subsequent calculations.
10. *See supra* note 2996. This figure adds the ongoing costs of disclosure associated with Governance (£9,100), Strategy (£13,000), and Risk Management (£10,800). The total (£32,900) is converted to USD based on the 2022 average exchange rate. (£32,900)*(1.2369 USD/GBP) = $40,694. This reflects a 32% reduction from initial to ongoing costs, which we consider in determining the appropriate percentage reduction in subsequent calculations.
11. *See supra* note 2987. This company reported that the cost of issuing its first TCFD report was less than $100,000. To be conservative, we assume $100,000 is the initial cost. To estimate ongoing costs, we refer to the percentage reduction from initial to ongoing costs as reflected by the FCA rule (56%) and the BEIS rule (32%), of which the median is 44%. ($100,000)*(1 – 0.44) = $56,000.
12. *See id.* This company reported that the cost of producing its first TCFD report was less than $10,000. To be conservative, we assume $10,000 is the initial cost. In estimating ongoing costs, we refer to the percentage reduction from initial to ongoing costs as reflected by the FCA rule (56%) and the BEIS rule (32%), of which the median is 44%. $10,000*(1 – 0.44) = $5,600.